

EEHC DISTRIBUTION MATERIALS SPECIFICATION

EDMS 24-100-2

Date: 21-03-2023

EDMS 24-100-2

TECHNICAL SPECIFICATION

FOR

GRID INTERCONNECTED

PHOTO VOLTAIC SYSTEM UP TO 30 MW DC

Issue: MAR-2023 / Rev- 2

▪ توجد بنود اختيارية (Option items) يجب تحديدها بواسطة شركة التوزيع قبل الطرح.

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1. ABBREVIATIONS

PV	PHOTOVOLTAIC
IEC	INTERNATIONAL ELECTROTECHNICAL COMMISSION
IEEE	INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS
NEMA	NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION
ASTM	AMERICAN SOCIETY FOR TESTING AND MATERIALS
UL	UNDERWRITER'S LABORATORIES
NFPA	NATIONAL FIRE PROTECTION ASSOCIATIONS
NEC	NATIONAL ELECTRICAL CODE
DSR	DISTRIBUTION SUBMITTAL REQUIREMENTS
PV	PHOTOVOLTAIC
STC	STANDARD TESTING CONDITIONS
GCR	GLOBAL COMPETITION REVIEW
PCC	POINT OF COMMON COUPLING
RCC	REINFORCED CEMENT CONCRETE
MPPT	MAXIMUM POWER POINT TRACKING
SPD	SURGE PROTECTION DEVICE
DAS	DATA ACQUISITION SYSTEM
ROQ	RECOMMENDED ORDER QUANTITY
BOM	BILL OF MATERIAL
PVIR	PV NOMINAL MAXIMUM POWER AT STC/INVERTER MAXIMUM AC OUTPUT POWER
UV	ULTRAVIOLET
PVDF	POLYVINYLIDENE FLUORIDE
PVF	POLYVINYL FLUORIDE
EVA	ETHYLENE VINYL ACETATE

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2. SCOPE

The work covered by this specification includes site survey, design, engineering, fabrication, testing, delivery to the site, installation (civil, mechanical, electrical...etc), supervision, training (if required), start-up, commissioning, and putting into successful operation for photovoltaic (PV) system.

The PV system shall have capacity up to 30 MW and have the electrical operation parameters as same as the local connected network.

The Photovoltaic system shall include, but not limited to, the following:

- A. Photovoltaic modules and array system
- B. Supporting structure and walkway including handrails
- C. Inverter(s)
- D. Inverter distribution panels
- E. Protection and metering system
- F. Data acquisition and analysis system
- G. Cables and raceways
- H. Surge suppression, earthing and bonding system

3. QUALITY STANDARDS

3.1 General

The quality of items and services supplied by Contractor shall be controlled to meet the requirements of this Specification, referenced codes and standards and other Contract Package documents.

3.2 Codes and Standards

- 3.2.1 The requirements of this Specification are based on the conformance with the following Egyptian Distribution, IEC, US Codes and standards as a priority as stated.
- 3.2.2 In case of conflict between performance or design characteristics given by the various codes and standards agencies that cover a common area of equipment and/or design and this Technical Specifications, the matter should be referred to the Distribution Company.
- 3.2.3 Referenced Codes and Standards

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The requirements of this specification are based on the conformance with the following Codes and standards:

Table (1)

S. No	Standard No.	Description
1.	IEC 60364-44	low voltage electrical insulations – Part 44 : Protection for safety - Protection against voltage disturbances and electromagnetic disturbances.
2.	IEC 60898-2	Circuit-breakers for overcurrent protection for household and similar installations - Part 2: Circuit-breakers for a.c. and d.c. operation.
3.	IEC 60904-1	Photovoltaic devices - Part 1: Measurement of photovoltaic current-voltage characteristics.
4.	IEC 60904-2	Photovoltaic devices - Part 2: Requirements for reference solar devices.
5.	IEC 60904-8	Photovoltaic devices - Part 8: Measurement of spectral response of a photovoltaic (PV) device.
6.	IEC 60947-2	Low-voltage switchgear and control gear - Part 2: Circuit-breakers.
7.	IEC 61215	Crystalline silicon terrestrial photovoltaic (PV) modules- Design qualification and type approval.
8.	IEC 61277	Terrestrial photovoltaic (PV) power generating systems - General and guide.
9.	IEC 61439	Low Voltage Switchgear and Control gear Assemblies.
10.	IEC 61643-1	Low voltage surge protective device –Part 1: Surge protective devices connected to low voltage power distribution systems- Requirements and tests.
11.	IEC 61683	Photovoltaic systems - Power conditioners - Procedure for measuring efficiency.
12.	IEC 61721	Susceptibility of a photovoltaic (PV) module to accidental impact damage (resistance to impact test).
13.	IEC 61724	Photovoltaic system performance monitoring - Guidelines for measurement, data exchange and analysis.

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14.	IEC 61727	Photovoltaic (PV) systems - Characteristics of the utility interface.
15.	IEC 61730	Requirements for construction, testing and safety qualification.
16.	IEC 61829	Crystalline silicon photovoltaic (PV) array - Onsite measurement of I-V characteristics.
17.	IEC 62093	Balance-of-system components for photovoltaic systems - Design qualification natural environments.
18.	IEC 62109-1	Safety of power converters for use in photovoltaic power systems - Part .1; General requirements.
19.	IEC 62109-2	Safety of power converters for use in photovoltaic power systems - Part 2: Particular requirements for inverters.
20.	IEC 61277	Terrestrial photovoltaic (PV) power generating systems - General and guide.
21.	IEC62116	Utility-interconnected photovoltaic inverters - Test procedure of islanding prevention measures.
22.	IEC 62446	Grid connected photovoltaic systems - Minimum requirements for system documentation, commissioning tests and inspection.
23.	IEC 62548	Design requirements for photovoltaic (PV) arrays.
24.	IEEE 928	Recommended criteria for terrestrial PV power systems.
25.	IEEE 1374	Guide for terrestrial PV power system safety.
26.	NEMAAB-1	Molded Case Circuit Breakers and Molded Case Switches.
27.	ASTM E1036	Standard Test Methods for Electrical Performance of Non-concentrator Terrestrial Photovoltaic Modules and Arrays Using Reference Cells.
28.	ASTM E1171	Test Methods for Photovoltaic Modules in Cyclic Temperature and Humidity Environments.
29.	ASTME 1799	Standard Practice for Visual Inspections of Photovoltaic Modules.
30.	ASTM E 1802	Standard Test Methods for Wet Insulation Integrity Testing of Photovoltaic Modules.
31.	ASTM E 1830	Standard Test Methods for Determining Mechanical Integrity of Photovoltaic Modules.

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32.	ASTM E 2047	Standard Test Method for Wet Insulation Integrity Testing of Photovoltaic Arrays.
33.	NFPA 70	Instrument Inverters, Converters, Controllers and Interconnection System Equipment for Use with Distributed Energy Resources.
34.	UL 1741	Flat-Plate Photovoltaic Modules and Panels.
35.	UL 1703	National Electrical Code.
36.	IEC TS 62738	Ground Mounted Photovoltaic Power Plants Design Guidelines and Recommendations.

Such regulations are available on the website:

www.egyptera.org/renwables or <http://www.cpa.gov.eg/electricity.htm>

also can be obtained from distribution company.

3.3 Contractor's Quality System Requirements

Contractor and his subcontractors, as applicable, shall have a quality management system that controls the inspection and testing of their product. Distribution Company or his representatives reserve the right to evaluate Contractor and subcontractors, quality management system documentation and to decide if the systems meet the requirements of the project. Distribution Company or his representatives also reserve the right to carry out appraisals and quality audits of Contractor and subcontractors, quality managements systems, at any time during the period of the Contract, to verify compliance with the contract.

4 SUBMITTALS [Appendix-A]

Contractor shall submit documents for Distribution Company's review in accordance with the Document Submittal Requirements (DSR) form.

5 SERVICE REQUIREMENTS

All equipment supplied shall have properly function under the site conditions, e.g. ambient air temperatures, humidity, and seismic loads, specified in "Technical Requirements for Connecting PV Systems to Low/Medium Voltage Distribution Networks" by Egyptian Electric Utility and Consumer Protection Regulatory Agency.

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6 DESIGN REQUIREMENTS

6.1 General

6.1.1 The Photovoltaic system shall be Grid connected to operate in parallel with the 400 VAC auxiliary system of the distribution company power plant to feed the grid. Means to prevent the reverse power to the PV plant or circulating current between the two sources are required.

6.1.2 The Photovoltaic system shall be designed to efficiently use the available space to generate the maximum output kWp DC at standard testing conditions STC and to generate the maximum allowable annual energy.

6.1.3 The contractor is responsible for site survey and Takes the necessary measurements to perform the calculation, and Photovoltaic system sizing and response for building safety against mechanical loads of the PV system.

6.1.4 The Contractor shall determine the most efficient mounting parameters of the PV arrays (Tilt, the orientation, GCR ratio, etc.) to maximize the solar energy exposure, and minimize the shade from the other equipment.

6.1.5 The PV arrays shall be uniformly arranged, and any cabling, raceways, junction boxes and other balance of system equipment shall be arranged and installed considering the access ways for maintenance requirements as recommended by the manufacturer.

6.1.6 The mounting of the PV arrays shall take into consideration the requirement of the maximum allowed air circulation for the PV modules natural cooling as required by the PV modules specifications.

6.1.7 For safety, The equipment specifications and the system installation shall meet the requirements of NEC articles 690 and 705.

6.1.8 The photovoltaic system arrangements shall be designed for easy maintenance.

6.1.9 PV power plants are often designed with high PVIR ratios to achieve lower installed cost of energy. From a technical standpoint, it is most important to consider the PVIR impact on the safe operation of the inverter. If the PVIR is too high the input rating of the inverter may be exceeded IEC TS 62738, it should be related to inverter ratings and site conditions.

6.1.10 performance ratio calculated by PV simulation program should be 80% at least.

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6.2 Photovoltaic modules and Array system

6.2.1 PV modules shall be according to IEC 61215, IEC 61730 .

6.2.2 PV modules shall contain crystalline silicon cells either mono or poly, where thin film or any other material shall be subjected to NAREA approval with plug and play water proof connectors, high transmission tempered glass, aluminum frame and adequate bypass diodes protection.

6.2.3 The connection to a module shall be arranged so that removal of a module or panel from a photovoltaic source circuit does not interrupt a grounded conductor to other PV source circuits.

6.2.4 PV arrays combiner junction box shall be provided for termination (required in case using 6.6.1 as the distribution company requirement).

6.2.5 The combiner box should have a provision for opening for replacing the cables.

6.2.6 PV modules shall be dust repel and capable of resisting damage/water penetration when subject to varying weather conditions including dust, sand, rain, washing fluids, and similar intermittent external conditions.

6.2.7 PV modules must be warranted for their output peak watt capacity, which should be with linear degradation not less than 90% at the end of 10 years and 80% at the end of 25 years starting from operating date of the PV system.

6.2.8 PV modules shall be operating and withstanding ambient temperature from -5°C to 60° C.

6.2.9 The performance warranty for PV modules is 25 years.

6.2.10 Tenderers must provide any distribution company clarifications.

6.2.11 The encapsulant used for the PV modules should be UV resistant in nature. No yellowing of the encapsulant with prolonged exposure shall occur. The sealant used for edge sealing of PV modules shall have excellent moisture IP with good electrical insulation and good adhesion strength.

Edge tapes for sealing are not allowed.

6.2.12 The back sheet of PV module shall be minimum of three layers with outer layer (exposure to ambience) and shall be made of PVDF or PVF. The Back sheets for PV Module with 2 layered or 3 layered Polyester types or the back sheets with Polyester (PET type) at Air side material are not permitted for the empanelment.

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The minimum thickness of the core layers (without adhesive and inner EVA coated). In case of bifacial modules, solar cells are equipped on both sides.

The maximum allowed water vapor transmission rate shall be less than 2 g / m²/day and shall have a Partial Discharge > / = 1500V DC.

6.3 Supporting structure and walkway including handrails

All of the following items related to Supporting structure and walkway including handrails concern Electricity Distribution Companies (EDC) only

6.3.1 There should preferably be a fixed arrangement (stand) for mounting the module in the direction facing the sun to obtain the maximum peak power output during year.

6.3.2 PV modules, module mounting frames, access walkway and the methods used for attaching frames to buildings shall be rated for the maximum wind speeds in the site.

6.3.3 Support structures and module mounting arrangements should comply with applicable building codes regulations and standards and module manufacturer's mounting requirements.

6.3.4 Provisions should be taken in the mounting arrangement of PV modules to allow for the maximum expansion/contraction of the modules under expected operating temperatures, according to the manufacturer's recommendations. Similar provisions should be taken for other applicable metallic components, including mounting structures, conduits and cable trunks. Module mounting frames, and the methods used for attaching modules to frames and frames to building shall be made from corrosion resistant materials suitable for the lifetime and duty of the system. The support structure used should be prefabricated hot dip galvanized steel (The minimum thickness of galvanization for hot dip Galvanized Mild Steel should be at least 85 microns) or treated aluminum 6063 or better (anodized, oxidized, etc). At Coastal Areas Ionized Aluminum frames Or Galvanized Mild Steel should be Used for installations. It does not allow any actions like punching or welding or paint in the site.

6.3.7 The PV module mounting structure shall have a capacity to withstand a wind velocity of 150 km/hr. Detailed specifications for the mounting structure are given below:

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Table (2)

Bolts, nuts, fasteners	Stainless steel SS 304
Mounting arrangement for RCC-flat	(Based on owner/ owner consultant approval) With removable concrete ballast made of pre-fabricated plain concrete cement (1:2:4) or with anchored roof mounting for RCC flat, ensuring that the roof remains water proof and ensuring stability and ability of withstanding mechanical stress resulting from winds.
Mounting arrangement for metal sheet	Mounting directly on the sheet metal, ensuring stability and wind withstanding capacity, or penetrating the sheet metal and fixing to the sub- structure, ensuring that the roof remains water proof and ensuring stability and wind withstanding capacity.
Mounting arrangement for elevated structures	The elevated structure has to be securely anchored to the supporting surface. Concrete foundations of appropriate weight and depth for elevated structures mounted directly on the ground; Bolted with anchor bolts of appropriate strength for elevated structures mounted on RCC surfaces.
Mounting arrangement for ground installations and pile based fixation	With RCC layer above pre-fabricated plain concrete cement (1:2:4) concrete ballast; assuring enough ground clearance to prevent damage of the module through water, animals and other environmental factors.
Minimum horizontal distance between roof barbet and mounting structure	0.6 m and depend on the site condition
Access for panel cleaning and maintenance	All solar panels must be accessible for cleaning and maintance.

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6.4 Inverter(s)

6.4.1 Inverter shall be in compliance with IEC-61727, IEC 62093, IEC 62116, and IEC 62109-1,2.

6.4.2 Contractor shall submit conversion efficiency data tested by an International Recognized Testing Laboratory, The conversion efficiency data is from testing on a minimum of five samples; at least 97% efficiency is required.

6.4.3 The inverter shall be equipped with the following:

- A. Input and output means, including disconnecting switches and sub combiners.
- B. Maximum Power Point Tracking (MPPT). (The Maximum Power Point tracking (MPP) range of the inverters should incorporate the MPP points of the arrays I-V curve at different temperatures).
- C. Anti-islanding features i.e the photovoltaic system in the event of voltage or frequency variations must island/disconnect itself with the time stipulated as per IEC standards (ON Grid/Hybrid) types. The Hybrid inverter should have all the technical requirements for connecting to the Grid, provision of Intentional Islanding with facility for connecting to a battery bank.
- D. The Hybrid inverter shall include appropriate self-protective and self-diagnostic feature to protect itself and the PV array from damage in the event of inverter component failure or from parameters beyond the inverter's safe operating range due to internal or external causes.
- E. IP65 enclosure for outdoor applications with sun shed and IP54 for indoor application.
- F. For each inverter there should be communication interface and providing a laptop (I5,6GB RAM, Disk 750 GB, Built in VEGA ATI) for each plant (option) with the corresponding programs installed on it.
- G. The Inverter shall be provided with built in UI monitor to display the electrical parameters or any suitable interface display. External monitor display the plant electrical data via remote communication system.

6.4.4 The operating range of the inverter should match the I-V characteristic of PV arrays.

6.4.5 Inverter should be provided with Arc fault circuit interruption function

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enabled.

- 6.4.6 The PV system output shall be three phase true sine wave with suitable voltage for the connection with the grid with maximum $\pm 10\%$ variation and 50 Hz frequency (48.5 – 51) Hz range of operation. The inverter maximum efficiency shall be at least 97% and the tenderer should indicate the loading corresponding efficiency range. The Total Harmonic distortion shall be limited to 5%. The injected direct current (DC) shall be limited to 0.5% of the alternating current nominal value.
- 6.4.7 Inverter shall be installed with sun shed provided by contractor in case of outdoor applications.
- 6.4.8 If the inverter protected with password to manage the software, then Distribution Company should be provided with this password.
- 6.4.9 Inverter shall be operating and withstanding ambient temperature from -5:60 degree Celsius.
- 6.4.10 The inverter warranty is 5 years.

After a voltage or frequency out-of-range condition that has caused the PV system to cease energizing the utility network, the PV system shall not re-energize the utility network for 1 min after the utility service voltage and frequency have recovered to within the specified ranges.

6.5 Inverter Distribution panels

- 6.5.1 Inverter distribution panels shall have ground fault protection on the main incoming breakers if not provided in inverter. Contractor shall be responsible for protection coordination with the downstream Contractor's breakers.
- 6.5.2 For 200 kW stations, the panel(s) shall be furnished with copper buses capable of momentarily withstanding the short circuit current stresses calculated by the Contractor in case of connecting more than one inverter.
- 6.5.3 For higher than 500KW stations, the PCC shall be at low/medium voltage, through suitable size step up transformers in case of medium voltage .
- 6.5.4 Enclosures shall be freestanding of metal, fully enclosed type with IP 65 for outdoor applications. Enclosures shall have adequate strength to withstand all stresses imposed by shipping, handling, installation, and

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operation without distortion or other damage.

6.6 Protection and metering system

- 6.6.1 A blocking diode shall be provided to block reverse flow of current into a photovoltaic source circuit (required according to distribution company requirements).
- 6.6.2 Suitable circuit breakers should be provided for isolating the PV system from the loads both in the DC side and the AC side.
- 6.6.3 The PV system shall be protected against overcurrent and short circuit by using suitable circuit breakers- these circuit breakers are certified to either IEC 60898-2 or IEC 60947-2.
- 6.6.4 The Contractor is responsible to set the protection setting point according to the equipment manufacturers' specifications.
- 6.6.5 Disconnecting means, residual current monitor, earth fault detection and alarm shall be provided.
- 6.6.6 The cables from the array strings to the solar grid inverters shall be provided with DC fuse protection (even internal DC Fuse available). Fuses shall have a voltage rating and current rating as required. The fuse shall be housed in thermoplastic IP65 enclosures with transparent covers.
- 6.6.7 A DC distribution box shall comprise the following components and cable terminations:
- Incoming positive and negative DC cables.
 - DC circuit breaker, 2 pole (the cables from the DC Combiner Box will be connected to this circuit breaker on the incoming side).
 - DC surge protection device (SPD), class 2 as per IEC 60364-5-53.
 - Outgoing positive and negative DC cables to the solar grid inverter.
- 6.6.8 Surge Protection
- Surge protection shall be provided on the DC side and the AC side of the solar system.
 - The DC surge protection devices (SPDs) shall be installed in the DC distribution box adjacent to the solar grid inverter.
 - The AC SPDs shall be installed in the AC distribution box adjacent to the solar grid inverter.
 - The SPDs earthing terminal shall be connected to earth through the

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above mentioned dedicated earthing system. The SPDs shall be of type 2 as per IEC 62305.

6.6.9 METERING and / smart meter

- Net meter: Net meter means the bidirectional energy meter to be installed at the interconnection point of the consumer with the network of distribution licensee for net metering and zero export controller device for self-consumption scheme.
- Solar Meter: A separate Energy Meter called Solar Meter shall be provided at the output of inverter or multiple inverters to record the energy generation from the Solar System. Solar energy meter means a unidirectional meter to be installed at the delivery point of the solar energy system to measure the solar electricity generated.

6.7 Data acquisition and monitoring system

6.7.1 The contractor shall furnish, install, commission and operate all measurements and a complete data acquisition and analysis system (using internet) to collect all measurements required to calculate the following High Level Metrics:

- A. Total PV System Production.
- B. PV System Standby Use.
- C. Equivalent Daily Hours of Peak Rated PV Production.
- D. Equivalent Annual Hours of Peak Rated PV Production.
- E. Total Incident solar radiation, wind speed, temperature (Optional).
- F. PV system AC Electricity Generation effectiveness.
- G. PV System Performance Ratio (Optional).
- H. Maximum Time-Series Net PV Production.
- I. Average Daily Time-Series PV Production Profiles.
- J. Avoided green house gas emissions due to the usage of the PV system.

6.7.2 Contractor shall provide the incident weather station.(Optional)

6.7.3 Contractor shall submit a list of measurements to be used and a description for the Web connecting device including all methods used for calculating each metric.

6.7.4 Measurements shall be on daily, monthly and annual basis.

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6.7.5 DAS shall be capable of generate reports for each metric.

6.8 Cables and raceways

- 6.8.1 DC Cables of PV system shall be in accordance with IEC 62548.
- 6.8.2 The overall voltage drop from the most remote module in the array to the terminals of the application circuit should not exceed 2% of the PV array voltage at its maximum power point. The overall voltage drop of AC cables should not exceed 5%.
- 6.8.3 For the DC cabling, XLPE or XLPO insulated and sheathed, UV stabilized single core flexible copper cables shall be used. Multi-core cables shall not be used. Voltage rating 660-1500.
- 6.8.4 Contractor shall design, furnish, install and test all power and control cables between Contractor's equipment/panels.
- 6.8.5 Contractor shall design, furnish and install all raceway systems (cable trunk, exposed conduits etc) for all cables and wiring furnished by the Contractor. All raceway system including cable trunks and conduits shall be hot dipped galvanized steel. (minimum height from ground level should be 10cm) .
- 6.8.6 Contractor shall design, furnish and install cable trunks for distribution company power cables that will feed power distribution panels mentioned in item 5.5 of this specification. Location of interface points with distribution company raceways will be electrical equipment room located in STG building ground floor of each unit.
- 6.8.7 Contractor shall design, furnish and install cable trunks for distribution company control cables that will feed DAS mentioned in item 5.8 of this specification. Location, of interface points will be the distribution company office. The above ground raceways shall be routed to facilitate access, troubleshooting, maintenance, etc.
- 6.8.8 The earthing conductor should be rated for 1.56 times the maximum short circuit current of the PV array.
- 6.8.9 The cross-section area of the earthing conductor from PV equipment to grounding system should be according to PV designed capacity and not less than 6 mm² if copper, 10 mm² if aluminium or 70 mm² if hot-dipped galvanized iron (for stations less than 10KW). For the earthing of lightning

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arrestor, cross-section of the earthing conductor should not be less than 16 mm² of copper or 70 mm² if hot-dipped galvanized iron.

6.8.10 To protect connection cables against UV sunrays and high temperature, the power outdoor cabling shall be made of copper conductors with double sheath type HO7RNF inside XLPE or PVC pipes or inside cable trunk.

6.9 Surge suppression, earthing and bonding system

6.9.1 Contractor shall be responsible to connect his equipment, inverter(s), inverter distribution panels, Supporting structure and walkway including handrails to main grounding grid and provide pigtails at locations of equipment.

6.9.2 The PV modules frames, array support structure, inverter electric boards, etc. shall all be earthed through grounding system to limit the resistance to 5 ohms. As well High voltage surge and lightning protection shall be supplied.

6.9.3 Bonding conductor shall be provided for functional or protective bonding.

6.9.4 The requirements of manufacturers of PV modules and manufacturers of inverter to which the PV array is connected shall be taken into account in determining the most appropriate system earthing arrangement.

6.9.5 For protection against electric shock, the requirements of IEC 60364 shall apply.

6.9.6 The PV system shall be protected against lightning surge and overvoltage. The surge arresters shall be type 2 (with reference to IEC 61643-1) rated at a continuous operating voltage of at least 125 percentage of the open-circuit voltage of the PV string, and a flash current of more than 5A.

7. SYSTEM POWER FACTOR

The overall system power factor shall be at least 0.92 according to distribution code. Measurements should be carried out at PCC (According to SSPV & MSPV Grid Code).

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8. MANUFACTURE

8.1 General Condition at Delivery

All equipment shall be free from rust, scale, manufacturing residue and foreign material to the extent that it can be put into operation without further cleaning.

8.1.1 All work performed at the jobsite on the equipment in connection with repair or replacement of defective materials or components shall be performed by Contractor or others under the supervision of the Contractor.

9. MARKING AND DOCUMENTATION

Marking and documentation shall be in accordance with IEC 62548. Modules shall be marked with identification of terminals as to polarity, maximum over current device rating for module protection, and with the following ratings:

- A. Open-circuit voltage
- B. Operating voltage
- C. Maximum permissible system voltage
- D. Operating current
- E. Short-circuit current
- F. Maximum power
- G. Temperature coefficients

10. SUPPLYING AND HANDLING

10.1 Contractor shall pack all equipment for supplying and prolonged storage according to the following requirements.

10.2 Preparation for delivery shall protect the PV system equipment and all other parts and devices or accessories against corrosion, dampness, breakage, or vibration injury during transportation and handling. Packing shall prevent tampering or pilfering and shall be approved and accepted by transportation companies.

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10.3 Contractor shall replace all equipment damaged due to improper preparation for transportation.

11. TESTING

11.1 PV modules Tests

11.1.1 PV modules must qualify enclose test reports according to IEC standard. Additionally the performance of PV modules at standard test conditions must be tested in accordance with IEC 61215 and IEC 62446.

11.1.2 Crystalline silicon photovoltaic (PV) array - On-site measurement of current voltage characteristics shall be in accordance with IEC 61829.

11.2 Test Reports

Test period of at least 3 months shall be carried out to ensure that actual KWH is in accordance with guaranteed KWH in condition of periodical maintenance. The Contractor shall-submit certified test data of all testing performed, in accordance with IEC 60904 Part 1 prior to shipment of the equipment. Testing shall be performed as required to ensure the proper and complete installation of all components. Required testing equipment shall be in good operating condition and shall be properly maintained and calibrated.

Upon completion of testing and checking of each item of equipment, any necessary maintenance and protection shall be carried out. Written documentation shall be maintained for all checking and testing results.

12. SITE TRAINING

If agreed the contractor shall make site training for the distribution company's technical staff on software, installation, operation, maintenance...etc for the supplied equipment.

13. DESIGN REVIEWS AND TIME SCHEDULE

The tenderer shall indicate his overall system design configuration and the Distribution company shall have the right to ask for and require any explanations. In addition the tenderer shall indicate in his offer his proposed implementation time schedule.

EEHC DISTRIBUTION MATERIALS SPECIFICATION

EDMS 24-100-2

Date: 21-03-2023

APPENDIX A

SUBMITTTLES

The following documents should be submit to the distribution company for example for review :

1. Certificate of renewable energy authority.
2. Systems Diagrams.
3. Spare parts list for commissioning and startup.
4. Electrical Single Line Diagram.
5. Structure design drawings and material specifications.
6. PV panels and structure installation drawings, indicate row Spacing .
7. Deviations to technical specification.
8. PV module and inverter data sheets.
9. PV array and inverter design calculations.
10. Preliminary ROQ, BOM and specifications inclusive of vendor names, model number.
11. Routine test report including modules serial numbers.
12. Estimated daily and yearly degradation of PV module power output .
13. Estimated plant energy generation in the first 2 year (KWH/year) calculated by one of PV simulation programs.
14. Total foot print area required for the plant .
15. Mechanical Consultant certificate to ensure that supporting structure should withstand environmental conditions (wind speed, temperature, moisture, etc.....) .
16. Break down voltage test report for PV modules.
17. Wet leakage test report for PV modules.